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THE IMMIGRATION ATTORNEYS

CAREER SUMMARY

Thank you for choosing Ellis Porter PLC's High-Skilled Immigration Team. We've built our practice on the belief that every case is unique and should be treated as such. We want to build a case strategy around your specific background, expertise, and credentials, and this is the document that will help us best accomplish that. We ask that you please fill out each section as completely and thoroughly as possible so that we can best understand your career and what makes you special. If you have any questions or concerns while you work on it, please let us know. We look forward to working with you to prepare the strongest petition possible.

Section A: Your Area of Expertise

In this section, we want to learn about the area in which you work. A few sentences should be sufficient to answer most of these questions.

1. **What do you consider to be your field? This should be a very short description – a few words at most – and should be broad enough to cover all of your past, current, and anticipated future work**:

   Cell Biology

2. **Please provide a brief summary of the focus of your past work**:

   My work focuses on understanding how cell-cell interactions can bring about changes in cell shape and cell content. In healthy circumstances, these cellular changes are required for the cell to accomplish its function/role. When cells are reshaped improperly, they cannot perform their proper function, and this is observed in many diseases (Alzheimer's disease, Schizophrenia, macular degeneration). My field of research involves characterizing how neighboring cells reshape each other, understanding how this neighboring activity goes wrong in diseased conditions, and identifying targets for treatment in cases linked to human diseases.

   During my PhD (September 2010-May 2017), I discovered that during the normal development in the organism C. elegans, germ cells are reshaped dramatically by neighboring gut cells. Germ cells are very important cells because they give rise to the sperm and egg; hence, their content (DNA, mitochondria, proteins, etc) are the sole components passed on to progeny. Specifically, large pieces of germ cells are "bitten off" (cannibalized) and removed by gut cells, drastically reducing their size and mitochondrial content. My research identified molecules required in the gut cells to accomplish cellular biting. Interestingly, previous works have shown that germ cells lose mitochondrial content during development, but it remained unknown how germ cells accomplish loss of mitochondrial content. In some instances, this reduction of germ cell mitochondrial content leads to debilitating mitochondrial diseases such as Pearson's syndrome. My research is the first evidence of a cellular mechanism that leads to germ cell loss of mitochondrial content.

During my Master's, I studied how neighboring cells contribute to the migration of precursor heart cells in sea squirts. This was an insightful model to study how the environment/neighborhood of cells can promote metastasis in cancer cells. In this study, I developed a tool to probe the contributing role of each neighboring cell type along the path of migration. This tool allowed us to block the secretory pathway of neighboring cells and show that these cells release factors that promote migration and guidance of precursor heart cells.

3. **Please provide a brief description of what you hope to do in your future work in the US**:

   In the future, my work will focus on how both neuron-neuron interaction and neuron-glia interaction lead to molecular and physiological changes in neurons, which in turn, leads to the formation of a memory. In order to form a memory, neurons do not act alone. They work together in a circuit or network to form memories. The basic rules of how a collection of neurons form a memory remains a mystery. Identifying these fundamental rules will unlock our understanding of how neurons form memories and how they can forget and lose memories. My work will identify molecules that promote memory formation and consolidation.

4. **Please provide a brief description of your expertise. What makes you an expert in your field? What separates you from your peers**?

   As a cell biologist, I have over 10 years of experience in visualizing and quantifying changes in cell shape and cell content. I am uniquely trained in the use of the cutting-edge imaging technique light-sheet microscopy, which allows for very fast imaging of live cells without killing or damaging the cells. This technique is extremely powerful and allows for imaging of cells in developing and moving animals without anaesthetization which could cause developmental defects. There are still very few scientists who have experience with this imaging technique. In the past, initial attempts at visualizing germ cell remodeling live with conventional confocal microscopy led to blurry illegible images because the embryo moves rapidly during development. To circumvent this problem, and for the first time in the field, I used light-sheet microscopy to image germ cell remodeling in live embryos. Also, after image acquisition, I have expertise in reconstructing images on computers and analyzing them. I also recreate these images of microscopic cell-cell interaction into detailed 3D models in order to better analyze these processes.

In addition, I have extensive experience in molecular genetics, specifically in the cutting-edge genome editing technique CRISPR. This has allowed me to target candidate molecules that control cellular processes of interest. Collectively, with my tremendous experience of handling stem cells (germ cells are the ultimate stem cells), I am uniquely placed above my peers with an arsenal of powerful skills to address biological questions. Most importantly, my extensive expertise in cell-cell interaction and cell behavior has allowed me to identify a *novel* cellular process (cell biting) and solve a big question in the field that was a mystery for over *30 years*.

5. **Please briefly describe the importance of work in your field. Broadly speaking, why is this type of work important**?

   Much of the breakthrough discoveries that led to development of effective therapeutics for human diseases began with studies in basic science research with model organisms such as C. elegans, mice and fruit-flies. For instance, a mutation in the gene Notch is the cause of many human leukemia cases. Notch was first identified and characterized in fruit-flies. Without this discovery, an understanding of how Notch and its mutations function would have been delayed for decades, if at all. Likewise, breakthrough genetic studies in C. elegans elucidated the mechanism underlying programmed cell death – a process that goes awry in cancer cells. Many discoveries in the molecular and cell biology of model organisms have accelerated development of therapeutics of human diseases. Specifically, working with simple organisms has allowed for the understanding of how molecules interact within a cell to make it function properly.

   Cell-cell interactions are an important and frequent occurrence in a living organism. For instance, photoreceptors (cells in the eye that let you see the world) shed pieces of themselves daily. Their neighboring cells, the retinal pigment epithelia (RPE) come in and clear off the shed pieces. When the RPE does not clear off the shed pieces in a timely manner, the photoreceptors degenerate. This is observed in the human genetic disease Best disease, where patients born with the mutation become blind as early as in their 20s. Mutations in a class of proteins called bestrophins lead to Best disease. In order to develop therapies to cure this disease before it leads to complete blindness, studies need to be done in simpler organisms to understand the function of these proteins during the cell-cell interaction of the RPE and photoreceptors. Likewise, cell-cell interactions are also important in the brain. Studies have shown that during development, neurons are reshaped at their synapses by neighboring glial cells. These reshaping or pruning events are important in rewiring the brain at key developmental stages such as adolescents. Currently, a growing collection of evidence suggest that schizophrenia might arise when these pruning events do not occur properly.

6. **Please briefly describe the impact of your specific work. Who benefits from the work that you do**?

I have worked to investigate mechanisms of specific cell-cell interactions that play critical roles in human diseases. This involves a detailed characterization of the key molecules participating in the mechanism using cutting edge genetic and molecular techniques such as CRISPR to either modify the molecules to create mutations similar to those found in human diseases or tag them with fluorescent markers so as to observe their action in live cells.

During my PhD, I discovered and characterized a new form of cell-cell interaction which I coined intercellular cannibalism or “cellular biting”. Though this cellular phenomenon was observed between germ cells and gut cells, there is compelling evidence that many other cell types might use this type of cell-cell interaction to reshape neighboring cells. For example, glial cells might use “cellular biting” to reshape neurons during development, suggesting that it might be a more widespread cellular phenomenon. Hence, my PhD work has direct impact on neuroscience, medicine and other related research foci.

In addition, my PhD work has greatly impacted the field of germ cell biology. Over 30 years ago, work by prominent scientists reported that C. elegans germ cells make protrusions called lobes. However, the fate and significance of these lobes remained a mystery in the field for decades. Using my unique and cutting-edge expertise, I was able to solve the mystery by showing that neighboring cells “bite off” and digest germ cell lobes in a spectacular process I coined intercellular cannibalism. I showed that this process dramatically reshapes germ cells by making them smaller and drastically reducing the mitochondrial content of the germ cells. My paradigm-shifting discovery (it was thought that lobes receded back into germ cells) has uncovered a critical point in germ cell development and is the first to show a cellular mechanism that leads to dramatic loss of mitochondrial content. My work has given a better understanding of germ cell development and revealed that germ cells can change their mitochondrial content, which will eventually give us a better understanding of the inheritance of the human mitochondrial diseases such as Pearson’s disease.

Section B: Your Academic Achievements

In this section, we want to get a detailed look at your educational history. Please answer the following questions for each degree you have earned. We realize some of this information may be in your CV, but it helps us work most efficiently to have all your information in one place.

1. **Degree earned**: BSc
2. **Area of study**: Biological Sciences
3. **Institution**: University of Illinois Urbana Champaign
4. **If known and notable, any rankings of your institution**: 53rd top national university (2019 US News and World Report)
5. **Dates of attendance**: Fall 2004 to Spring 2008
6. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:
    - Midwestern Scholar 2006 and 2007

7. **Degree earned**: MSc
8. **Area of study**: Molecular Biology
9. **Institution**: Columbia University
10. **If known and notable, any rankings of your institution**: 45th top national university (2019 US News and World Report)
11. **Dates of attendance**: Fall 2008 to Spring 2010
12. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:
    - 1st Place Presentation Prize at 2010 Columbia Biology Seminar

13. **Degree earned**: PhD in Cellular Biology
14. **Area of study**: Genetics
15. **Institution**: Boston University
16. **If known and notable, any rankings of your institution**: 5th Best Northeastern US School in Research (2019 US News and World Report).
17. **Dates of attendance**: Fall 2010 to Spring 2017
18. **Please list your accomplishments while pursuing this degree. For instance, honors, scholarships, student awards, etc. can be listed here**:
    - Janice and Karl Klapper Memorial Scholarship Fellow 2011- 2014
    - 1st Place Best Poster Presentation, 2015 Society of Genetics Research (SGR) International Meeting

Section C: Your Employment History

In this section, we want to get a detailed look at your employment history. Please answer the following questions for each full-time position in your field that you have held. You do not need to include employment from before receiving your Bachelor's degree. We realize some of this information may be in your CV, but it helps us work most efficiently to have all your information in one place.

1. **Employer name**: University of California, Los Angeles
19. **If known and notable, the reputation of your employer**: 21st top national university (2019 US News and World Report)
2. **Job title**: Research Assistant
3. **Detailed job description and list of duties**: Conducting and supervising research on the role of cell-cell interactions on memory formation.
4. **Dates of employment**: August 2017 through present
5. **Salary**: $48,000/year (2019)

Section D: Detailed Summary of Your Work

This is perhaps the most important section in allowing us to understand your work. Here, you should break your career up into different "projects" and provide a description of what you did. A project could be one specific area on which you worked for an extended period of time, a collection of smaller tasks that were related in some way, or a specific step in your career progression. Please try to describe at least 2-3 projects, but no more than 6. In this template, we have included spots for 3 projects; please add additional entries as needed. If you're having trouble deciding how to organize or select your projects, please let us know and we can provide some suggestions about what might be best for you.

1. **PROJECT 1 – Name**: Study of the migration of precursor heart cells in *Ciona intestinalis*
2. **Date project began**: 09/2008 (Columbia University)
3. **Date project ended**: 03/2010 (Columbia University)
4. **Please list any funding this project received (if applicable)**:

    - National Science Foundation (NSF) [Grant #] [Award Amount]

5. **Please list all resulting publications (if applicable)**:

    Panter, K., Archibald, N., Luce, B., Bagcroft, H., Johnson, P. **Analysis of the Impact of Cell Type on the Behavior of Migratory Cells.** Cell Metabolism. 114, 218-226 (2010).

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish**?

    Cell migration is a key aspect of embryonic development. In many cases, after cells are born, they migrate long distances through neighborhoods of different cell types until they arrive at their destination where they will perform their function. Along the way, surrounding cells along the path of migration contribute to the migratory cues and directionality by releasing several molecules into the extracellular space. For instance, soon after the birth of germ cell precursors, they migrate through great distances of ever-changing landscapes of cells in the developing human embryo. In some cases, germ cells lose their way during migration, and if not killed, can become the source of germ cell tumors. Hence, understanding what molecules are released along the path of migration and how these molecules are interpreted by the migrating germ cells is pertinent to preventing germ cell tumors. In addition, cancerous cells can take advantage of these molecular migratory cues and become metastatic, migrating to other parts of the body. Hence, studying cell migration is an important line of inquiry in developing therapeutics to prevent metastasis during cancer. A big problem in cell migration is the complexity of migratory path, the long distances migrating cells take to reach their destination, and the variability from animal to animal in the exact path the cells take. In order to circumvent this problem, we employed a simple system, *Ciona intestinalis* embryos, where two precursor heart cells (TVCs) collectively migrate invariantly through a single path and relatively short distance to their destination. Therefore, in every embryo, TVCs migrate through the same group of cells. This feature of the simple system allows querying of every cell type along the path of migration on their contribution to the migratory behavior of TVCs. Also, because the embryos are transparent and amenable to long term imaging, we were able to watch the migration in live embryos. Our goal was to identify whether every cell type along the migratory path contributes to the successful migration of TVCs.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

   The aim of this project was to understand whether surrounding cells contribute to the invariant migratory behavior of the precursor heart cells (TVCs). In order to do this we first needed to fully characterize the normal migration of TVCs in live embryos. We accomplished this by generating transgenic embryos (embryos where foreign DNA has been introduced). These transgenic embryos had been modified so that TVCs are marked with a fluorescent protein (so that we can see the otherwise transparent cells) and the surrounding cells are marked with another fluorescent protein. This modification of the embryos allowed us to watch the TVCs as they migrate through surrounding cells in live developing embryos, which in turn, allowed for the full characterization of the migratory behavior.

   Next, we developed a system to prevent cells along the migratory path from releasing molecules into the extracellular space. To accomplish this manipulation, we decided on specifically inhibiting molecules in the secretory pathway of cells using genetic tools. We engineered an inhibition system where we expressed an abundance of an inhibitor that would halt the release of molecules out of the cell specifically in our cells of interest. After validating the inhibition, we systematically inhibited the secretory pathway of different cell types (epidermis, endoderm, mesenchyme, and notochord) along the path of migration. In each inhibitory experiment, we watched and documented how the manipulation affected the migratory behavior of TVCs.

   Overall, this study showed that different cell types can have distinct contribution/roles to the migratory behavior of cells. It reveals the complexity of migrating cells and how they have to interpret multiple inputs in their environment to guide their way to their destination. Similarly, cancer cells can take advantage of multiple inputs in the environment to become metastatic, revealing the need of a multipronged approach to preventing metastasis in cancer.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

   My primary role in this project was to design the tools to inhibit secretion of neighboring cells along the migratory path. This was a critical role in the project because without an inhibitory tool, we would not have been able to query the function of the neighboring cells in the migration of the TVCs. I assisted in conceiving the goals of the project and in designing of experiments.

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

   This project resulted in a published research article in a highly accredited journal *Cell Metabolism* (Impact Factor 4.123). For the field, it is a well cited paper and the findings revealed how different cell types along a migratory path can distinctly contribute to the migratory pattern of cells, and therefore illuminating the multifaceted approach one must take to prevent metastasis of cancer cells. This work was also highlighted in *Faculty 1000* as a "New finding" in the field of Cellular Biology.

1. **PROJECT 2 – Name**: Description of the developmental remodeling of C. elegans germ cells by cell cannibalism.
2. **Date project began**: 2011
3. **Date project ended**: 2017
4. **Please list any funding this project received (if applicable)**:

   - National Institutes of Health (NIH) [Grant #] [Amount]
   - Massachuseets Science Fund (MSF)

5. **Please list all resulting publications (if applicable)**:

   Bagcroft, H., Vance, K., Masters, L. **Impact of the modification of germ cells on digestive cellular health.** *Cell Stem Cell.* 12, 1485-1501 (2015).

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish**?

   Germ cells are extremely important cells to an organism. They are the cells that give rise to the sperm and egg, hence these are the cells that allow an organism to reproduce. Without germ cells, an animal will be sterile. For this reason, germ cells are considered the ultimate stem cells because one can make a whole organism from these cells. During embryonic development, germ cells form early and are set aside from all other cells. They undergo unique molecular and cellular remodeling to preserve their stem cell fate. When germ cells develop improperly, the animal ends up sterile and unable to pass on its genes to the next generation. In some cases, improper germ cell development leads to germ cell tumors in the adult animal. A conserved but poorly understood aspect of germ cell development is their intimate interaction with endodermal cells (gut cells). Midway through the development of *C. elegans* embryos, germ cells form large protrusions called lobes that become embedded inside the adjacent gut cells. By the end of embryonic development, lobes are no longer present. It was assumed that lobes become reabsorbed back into germ cells, but no such evidence was put forward. Hence, the fate of the lobes and the significance of germ cell-gut cell interaction has remained a mystery for decades. The reason why this remained a mystery is because it was a hard question to address. Embryos begin moving rapidly during the stages when germ cells interact with gut cells. Any attempt at imaging and documenting the interaction led to blurry images that could not be analyzed. Therefore, my goals were:

   - To solve the decades old mystery of the fate of germ cell lobes in live embryos using cutting edge light sheet microscopy imaging.

- To finally understand the relevance of germ cell-gut cell interaction.
- To discover and characterize the molecules needed for germ cell-gut cell interaction.
- To address whether germ cell content is being modified/reshaped during cell-cell interaction with gut cells and if so, how the modification affects germ cell development.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

 Goal 1: First, in order to solve the decades old mystery, I generated transgenic embryos where germ cells were labeled on their membranes exclusively by one fluorescent protein (red) and gut cells were labeled on their membranes by another fluorescent protein (green). This allowed visualization of the outlines of both cell types in an otherwise transparent embryo. Using light sheet microscopy, I was able to observe that fragments of germ cell membrane (labeled with red fluorescent protein) were inside green gut cells, which suggested that gut cells "eat" germ cell lobes. With light sheet microscopy, I could detail the steps in the eating process in live, moving embryos. This was a major discovery because it revealed that germ cells were being reshaped in size during development – a feature of germ cell development that was previously unknown. It was also a major discovery because for the first time, I presented clear evidence that lobes are not reabsorbed back into germ cells. Rather, lobes are removed from germ cells, dramatically reducing their volume. To confirm this, I reconstructed the images in 3D using computer software and measured the volume of the germ cells during development. Indeed, this revealed that germ cells have a drastic reduction in volume (losing 2/3 of their volume).

 Goal 2: After discovering germ cells are reshaped in size during development, I next wanted to understand whether gut cells played an active role in reshaping germ cells. One could imagine that germ cells can shed off pieces of themselves on their own without the need of gut cells. To address this, I looked into embryos lacking gut cells using a mutant identified in a previous study. Surprisingly, I found that in these mutants, lobes remain present or persist and are not removed, indicating that gut cells are required to remove lobes. In other to further understand the extent to which gut cells are involved in removing lobes, I asked whether persistent lobes in mutant embryos lacking gut cells remained connected to the germ cells. If so, it would mean germ cells remain connected to lobes by a small cytoplasmic bridge that is difficult to detect without super-resolution microscopy. This was an important question to ask because its answer determined whether gut cells were "biting off" lobes or simply eating lobes that were being fed to them. An analogous example is the contrast between plucking apples still attached to a tree (lobes are apples; germ cells are trees) or picking apples that have fallen off a tree. To distinguish between these two possibilities, I observed the diffusion of a dye from germ cells to lobes. If germ cells and lobes remain connected in mutant animals lacking gut cells, then the dye would be able to diffuse from germ cells into lobes. And if germ cells and lobes are not connected in mutant animals lacking gut cells, then the dye will not be able to diffuse from germ cells into lobes. The experiment revealed that in mutant embryos lacking gut cells, dye can diffuse from germ cells to lobes, indicating that germ cells and lobes remain connected by a cytoplasmic bridge when gut cells are absent. Ultimately, this was an exciting discovery because it showed that gut cells have an active role in eating lobes and that they do so by "biting off" lobes from germ cells. With this experiment, I discovered a new type of cell biological process I coined "intercellular cannibalism" (or cellular biting).

Goal 3: Next, I wanted to uncover the molecules that control cellular biting and characterize their function in the process. Using molecular and genetic tools, including CRISPR, I discovered and characterized four molecules required for intercellular cannibalism. These molecules are the GTPase Rac, actin, dynamin and the sorting nexin SNX9. My genetic experiments showed that all four molecules are required in gut cells to facilitate intercellular cannibalism. To figure where these molecules are acting inside gut cells, I labeled each molecule with fluorescent proteins and used light sheet microscopy to image the localization of these molecules in gut cells during intercellular cannibalism in live, moving embryos. I found that actin, dynamin and SNX9 all localize specifically to the membranes of the gut cells that touch lobe necks (the area between lobes and germ cells). At these sites, these molecules form a molecular machinery that acts like a noose on a rope where they tighten the cell membranes until they break. We call this machinery a scission complex. My experiments showed that actin localization to lobe necks is dependent on both Rac and SNX9 while the localization of dynamin was solely dependent on SNX9.

Goal 4: In other to determine whether intercellular cannibalism reshapes or modifies germ cell content, I looked into the content of germ cells. Surprisingly, I found that due to intercellular cannibalism, germ cells lose majority of their mitochondria. Mitochondria is an essential organelle that generates energy for a cell. A cell cannot exist without mitochondria. Interestingly, mitochondria also contain their own DNA which can develop mutations that cause mitochondrial diseases such as Pearson's disease. Many studies have shown that mitochondrial diseases can be inherited because germ cells lose mitochondrial content. The cellular basis of how germ cells lose their mitochondrial content has remained unknown and is a big question in the field. My experiments showed that majority of germ cell mitochondria are moved into lobes prior to intercellular cannibalism where they are eliminated along with lobes. The few remaining mitochondria left behind replicate and become the mitochondria that are passed on to the next generation of progeny. In mutants where lobes fail to form or fail to be removed efficiently, germ cells end up with a lot more mitochondria, revealing a possible mechanism in preventing mitochondrial diseases.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

   I was the primary researcher in this project. I was responsible for conceiving the project and the individual experiments within it along with my supervisor. I performed and analyzed 90% of the experiments in this project. My analysis was critical in understanding the relevant results which made the breakthrough discoveries. Without my initial observations, this project would not exist. My supervisor and I wrote the resulting research article that was published in a high impact journal.

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

   We ended up publishing my findings in the journal of Cell Stem Cell which has a high impact factor of 8.683 as of 2015 when the paper was published. Several review articles were written about my work in scientific journals by independent leading experts of international scientists in the field:

   - Lurk, J.S. & Kumar, V. **The impacts of intercellular cannibalism on cellular health.** *Cell Biol.* (2017).
   - Billings, L & Marche, K. **Cellular Cannibalism and Development.** *Cell* (2017).

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1. **<u>PROJECT 3</u> – Name**: Understanding synaptic plasticity through the use of modular neuronal circuits.
2. **Date project began**: August 2017
3. **Date project ended**: Ongoing
4. **Please list any funding this project received (if applicable)**: James Lazak Research Fund
5. **Please list all resulting publications (if applicable)**: No publication yet

6. **Please provide any necessary background information for us to understand this project. What is the significance of this work? Why did you undertake this project? Was there a specific issue or problem you were trying to resolve? What were you hoping to accomplish**?

   Neurons do not act on their own to form memories. A neuron must work with a collection of other neurons it interacts with at contact sites called synapses. We call these collection of neurons neuronal networks or circuits. In addition, neurons interact with neighboring cells called glia that also contact synapses. Glia are known to modulate how neurons interact and signal to each other. Together, these cell-cell interactions work to encode memory in the brain. To retrieve a memory, groups of neurons fire together in the same pattern that was created from the original experience. How these neurons become primed to fire in unique patterns which encode memory is still a major question in biology. Many studies have worked out clues, but because of the complexity of neuronal networks and the brain, a holistic and complete understanding remains elusive. In order to bypass this problem of complexity, I am building my own customizable circuits using cultured mice neurons. From this system, I will be able to stimulate neurons within a circuit for long periods of time and observe in real-time how their pattern of firing changes over time. The goals of this project are:

   - To build a culture system of customized network of stem-cell derived neurons.
   - To observe how patterns of firing change with a systematic program of stimulation of neurons.
   - To characterize a basic guiding rule all neurons follow to encode memory in the brain.
   - To identify molecules that promote memory formation in networks of neurons.

   Elucidating the basic rules neuronal networks use to encode memory will help us identify molecules that promote long-term memory formation. Findings from this study will also provide clinical avenues that will prevent or delay the effects of such neurological disorders such as Alzheimer's disease and dementia.

7. **Please provide a plain language summary of what was done in this project. You should try to be detailed (2-4 paragraphs is appropriate), as more information will help us better understand the work**:

Goal 1: In order to build a culture system of customizable network of mouse neurons, I first determined that I must have an unlimited source of neurons to experiment on. Using primary cell culture (neurons harvested from whole animal) seemed unrealistic because it would require continuously sacrificing mice to harvest their neurons. This is very expensive and labor intensive. Instead, I decided to derive neurons from mouse stem cells. Taking this approach is more advantageous for the following reasons. First, stem cells, when maintained appropriately, can continuously divide and hence provide an unlimited source of cells. Second, one can robustly derive neurons from stem cells. Third, one can modify the genome of stem cells so that they express transgenes. For these reasons, I generated stem cells from mice that have been engineered to express channel rhodopsin. With channel rhodopsin, neurons can be stimulated to fire using light.

In order to customize simple neuronal networks or circuits, I employ the technique of micropatterning. This is a very specialized and advance technique in studying cell biological processes. In the field, there are still very few scientists with experiences in micropatterning. The technique uses photolithography to create miniature (scale of microns) patterns on substrates that allow precise control of cell adhesion, growth and guidance. Using micropatterning, I customize neuronal networks (the pattern to which neurons interact with one another or make synapses with one another).

Goals 2&3: Using light sheet microscopy, I image and stimulate customized neuronal networks over long periods of time. My lab at Rockefeller U. has its own home-built light sheet microscope. This is such a unique advantage because very few labs can boast of having their own home-built light sheet microscope. A light sheet microscope is most advantageous for long term imaging of live culture cells because there is little to no phototoxicity introduced into cells during imaging and so cells can be imaged for a long time (several days) without any adverse effects. As mentioned earlier, my experimental neurons have been modified to express channel rhodopsin – a molecule that incites them to fire when exposed to light of certain wavelength. Using programmable software, I design a stimulation protocol where the microscope is directed to stimulate neurons at defined intervals. The goal is to observe how the pattern of firing changes after an extended period of stimulation. When neurons fire, there is a transient increase in calcium ions and this can be visualized using dyes that bind to calcium ions. With this experiment, I will be able to define general basic rules neurons follow to form a pattern of activation, which in turn, encodes a memory.

Goals 4: In the future, I plan to identify molecules that promote the formation of memory. For this endeavor, I plan to use a CRISPR screen where I cause mutations in my experimental stem cells and ask how these mutations affect memory formation. Mutants that have defects in memory formation will be mutated for candidate molecules that promote memory formation. In contrast, mutants that have an enhancement in memory formation will be mutated for candidate molecules that repress memory formation.

8. **What was your specific role on this project? What were you individually responsible for? How was your work critical to the outcome of the project**?

   I am the lead scientist in this project. I am responsible for conceiving the concepts and experiments in this project with the help of my supervisor. I design, perform and analyze 100% of the experiments. My expertise on stem cells, light sheet imaging, micropatterning and cell-cell interactions makes me uniquely suited for this groundbreaking research project.

9. **How was the project successful? What kind of impact has it had? Has this work been recognized for its significance, such as with awards or articles written about it? Has this work been used by other individuals or organizations, or changed others' practices in some way? Please provide specific examples where possible**.

   The project is still under development. In the near future, the results of this project will garner a lot of global attention as understanding how memory forms is a biological question that is of great interest to neuroscientists and the medical field.

Section E: Professional Activities

This section is focused on activities outside of your educational and employment history. It can involve selective memberships or leadership positions with professional organizations, peer review and editorial board experience, work organizing professional conferences or seminars, etc. Try to think about any roles for which you were chosen based on the significance of your past work or your expertise in the field.

Review experience (5):

1. Molecular Cell – 3 papers
2. Nature Reviews Genetics – 2 papers